

Math 241, F04
Exam 1

Name:
SSN: xxx-xx-

Instructions:

- There are totally 6 problems.
- You must show all of your work to receive a credit for a correct answer.
- No graphical calculator is allowed in the quiz and exam.

Problem	Points	Score
1	15	
2	20	
3	15	
4	20	
5	15	
6	15	
Total	100	

Good Luck !

Problem 1. (15 points) Given parametric equations of a curve in the plane:

$$x = 3 \sin t, \quad y = 4 \cos t, \quad 0 \leq t \leq \pi/4$$

a) Eliminate the parameter to obtain the corresponding Cartesian equation.

Solution: Since

$$\sin t = \frac{x}{3}, \quad \cos t = \frac{y}{4}$$

We get

$$\begin{aligned} \left(\frac{x}{3}\right)^2 + \left(\frac{y}{4}\right)^2 &= \sin^2 t + \cos^2 t \\ \frac{x^2}{9} + \frac{y^2}{16} &= 1 \end{aligned}$$

b) Find dy/dx without eliminating the parameter.

Solution:

$$\begin{aligned} \frac{dx}{dt} &= 3 \cos t \\ \frac{dy}{dt} &= -4 \sin t \\ \frac{dy}{dx} &= \frac{dy/dt}{dx/dt} = \frac{-4 \sin t}{3 \cos t} = -\frac{4}{3} \tan t \end{aligned}$$

Problem 2. (20 points) Let $\mathbf{a} = \langle 2, -5, -1 \rangle$, $\mathbf{b} = \langle 3, -1, 0 \rangle$ and $\mathbf{c} = \langle 1, 0, -6 \rangle$, find each of following:

a) $|\mathbf{a}|$, $|\mathbf{c}|$, and $\mathbf{a} \cdot \mathbf{c}$

Solution:

$$|\mathbf{a}| = \sqrt{2^2 + (-5)^2 + (-1)^2} = \sqrt{30}$$

$$|\mathbf{c}| = \sqrt{1^2 + 0^2 + (-6)^2} = \sqrt{37}$$

$$\mathbf{a} \cdot \mathbf{c} = 2 \cdot 1 + (-5) \cdot 0 + (-1) \cdot (-6) = 8$$

b) The cosine of the angle between $\mathbf{a}$ and $\mathbf{c}$.

Solution:

$$\cos \theta = \frac{\mathbf{a} \cdot \mathbf{c}}{|\mathbf{a}||\mathbf{c}|} = \frac{8}{\sqrt{30}\sqrt{37}} \approx 0.2401$$

c) $\mathbf{b} \times \mathbf{c}$

Solution:

$$\begin{aligned} \mathbf{b} \times \mathbf{c} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 3 & -1 & 0 \\ 1 & 0 & -6 \end{vmatrix} \\ &= \begin{vmatrix} -1 & 0 \\ 0 & -6 \end{vmatrix} \mathbf{i} - \begin{vmatrix} 3 & 0 \\ 1 & -6 \end{vmatrix} \mathbf{j} + \begin{vmatrix} 3 & -1 \\ 1 & 0 \end{vmatrix} \mathbf{k} \\ &= 6\mathbf{i} + 18\mathbf{j} + \mathbf{k} \end{aligned}$$

Problem 3. (15 points) Given the position vector $\mathbf{r}(t) = t\mathbf{i} + t^2\mathbf{j} + 3t^3\mathbf{k}$,

a) Find the velocity $\mathbf{v}(t)$, speed $|\mathbf{v}(t)|$ and acceleration $\mathbf{a}(t)$.

Solution:

$$\begin{aligned}\mathbf{v}(t) &= \mathbf{r}'(t) = i + 2t\mathbf{j} + 9t^2\mathbf{k} \\ |\mathbf{v}(t)| &= \sqrt{1^2 + (2t)^2 + (9t^2)^2} = \sqrt{1 + 4t^2 + 81t^4} \\ \mathbf{a}(t) &= \mathbf{r}''(t) = 2\mathbf{j} + 18t\mathbf{k}\end{aligned}$$

b) Find the symmetric equation of the tangent line to the curve at the point $P(-1) = (-1, 1, -3)$.

Solution: At $t = -1$, we have

$$\mathbf{v}(-1) = \mathbf{i} - 2\mathbf{j} + 9\mathbf{k}$$

So $\langle 1, -2, 9 \rangle$ is the direction of the tangent line to the curve at $P(-1) = (-1, 1, -3)$, then its equation is

$$\frac{x+1}{1} = \frac{y-1}{-2} = \frac{z+3}{9}$$

Problem 4. (20 points)

- a) Find the equation of the plane perpendicular to the vector $\langle 2, -1, 3 \rangle$ at the point $(1, 0, 4)$.

Solution:

$$2(x - 1) - y + 3(z - 4) = 0$$

$$2x - y + 3z = 14$$

- b) Find the parametric equations for the line through $(6, 1, -3)$ and $(-2, 2, 3)$.

Solution: The direction vector for the line is

$$\langle -2, 2, 3 \rangle - \langle 6, 1, -3 \rangle = \langle -8, 1, 6 \rangle$$

So the parametric equations for the line are

$$x = -2 - 8t, \quad y = 2 + t, \quad z = 3 + 6t$$

(* The answer is not unique)

- c) Find the center and radius of the sphere with equation

$$x^2 + y^2 + z^2 + 6x - 2y + 5 = 0.$$

Solution: Complete the squares:

$$x^2 + 6x + y^2 - 2y + z^2 = -5$$

$$x^2 + 6x + 9 + y^2 - 2y + 1 + z^2 = -5 + 9 + 1$$

$$(x + 3)^2 + (y - 1)^2 + z^2 = (\sqrt{5})^2$$

$$\text{Center: } (-3, 1, 0), \quad \text{Radius: } \sqrt{5}$$

Problem 5. (15 points) Consider the curve $\mathbf{r}(t) = e^t \sin t \mathbf{i} + e^t \cos t \mathbf{j} + e^t \mathbf{k}$, $1 \leq t \leq 3$, find each of the following:

a) $\mathbf{r}'(t)$ and $|\mathbf{r}'(t)|$

Solution:

$$\begin{aligned}\mathbf{r}'(t) &= e^t(\cos t + \sin t)\mathbf{i} + e^t(\cos t - \sin t)\mathbf{j} + e^t\mathbf{k} \\ |\mathbf{r}'(t)| &= \sqrt{e^{2t}(\cos t + \sin t)^2 + e^{2t}(\cos t - \sin t)^2 + e^{2t}} \\ &= \sqrt{e^{2t}(2\cos^2 t + 2\sin^2 t + 1)} \\ &= \sqrt{3}e^t\end{aligned}$$

b) Find the length of curve from $t = 1$ to 3.

Solution:

$$\begin{aligned}s &= \int_1^3 |\mathbf{r}'(t)| dt \\ &= \int_1^3 \sqrt{3}e^t dt \\ &= \sqrt{3}e^t \Big|_1^3 \\ &= \sqrt{3}(e^3 - e^1) \approx 30.0810\end{aligned}$$

c) The unit tangent vector $\mathbf{T}(t)$ at $t = \frac{\pi}{4}$.

Solution:

$$\begin{aligned}\mathbf{r}'\left(\frac{\pi}{4}\right) &= e^{\frac{\pi}{4}}\left(\frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}\right)\mathbf{i} + e^{\frac{\pi}{4}}\left(\frac{\sqrt{2}}{2} - \frac{\sqrt{2}}{2}\right)\mathbf{j} + e^{\frac{\pi}{4}}\mathbf{k} \\ &= \sqrt{2}e^{\frac{\pi}{4}}\mathbf{i} + e^{\frac{\pi}{4}}\mathbf{k} \\ |\mathbf{r}'\left(\frac{\pi}{4}\right)| &= \sqrt{3}e^{\frac{\pi}{4}}\end{aligned}$$

So

$$\mathbf{T}\left(\frac{\pi}{4}\right) = \frac{\mathbf{r}'\left(\frac{\pi}{4}\right)}{|\mathbf{r}'\left(\frac{\pi}{4}\right)|} = \frac{\sqrt{2}e^{\frac{\pi}{4}}\mathbf{i} + e^{\frac{\pi}{4}}\mathbf{k}}{\sqrt{3}e^{\frac{\pi}{4}}} = \sqrt{\frac{2}{3}}\mathbf{i} + \sqrt{\frac{1}{3}}\mathbf{k}$$

Problem 6. (15 points)

a) Change the **spherical** coordinate $(4, \pi/3, \pi/6)$ to Cartesian coordinates.

Solution:

$$\begin{aligned}x &= \rho \sin \phi \cos \theta = 4 \sin \frac{\pi}{6} \cos \frac{\pi}{3} = 1 \\y &= \rho \sin \phi \sin \theta = 4 \sin \frac{\pi}{6} \sin \frac{\pi}{3} = \sqrt{3} \\z &= \rho \cos \phi = 4 \cos \frac{\pi}{6} = 2\sqrt{3}\end{aligned}$$

So the Cartesian coordinates are

$$(1, \sqrt{3}, 2\sqrt{3})$$

b) Change the Cartesian equation $x^2 + y^2 + 6z^2 = 10$ to **cylindrical** coordinate equation.

Solution: We know

$$x = r \cos \theta, \quad y = r \sin \theta, \quad z = z$$

So we have

$$\begin{aligned}r^2 \cos^2 \theta + r^2 \sin^2 \theta + 6z^2 &= 10 \\r^2 + 6z^2 &= 10\end{aligned}$$